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# Equalizing an RGB Image, equalizing only L channel in Lab color space (2107042)
input_img = cv2.imread("boat.jpg")
reference_img = cv2.imread("power_plant.jpg")

# BGR to RGB, Then RGB to LAB
input_rgb = cv2.cvtColor(input_img, cv2.COLOR_BGR2RGB)
reference_rgb = cv2.cvtColor(reference_img, cv2.COLOR_BGR2RGB)
input_lab = cv2.cvtColor(input_rgb, cv2.COLOR_RGB2LAB)
reference_lab = cv2.cvtColor(reference_rgb, cv2.COLOR_RGB2LAB)

# Only take the L channel
input_L = input_lab[:, :, 0]
reference_L = reference_lab[:, :, 0]

# Calculate PDFs
input_hist = np.bincount(input_L.ravel(), minlength=256)
reference_hist = np.bincount(reference_L.ravel(), minlength=256)
input_pdf = input_hist / input_L.size
reference_pdf = reference_hist / reference_L.size

# Calculate CDFs
input_cdf = np.cumsum(input_pdf)
reference_cdf = np.cumsum(reference_pdf)

# Histogram matching

mapping = np.zeros(256, dtype=np.uint8)

for r in range(256):
    difference = np.abs(reference_cdf - input_cdf[r])
    z = np.argmin(difference)
    mapping[r] = z

output_L = mapping[input_L]
output_lab = input_lab.copy()
output_lab[:, :, 0] = output_L

# Back to RGB
output_rgb = cv2.cvtColor(output_lab, cv2.COLOR_LAB2RGB)

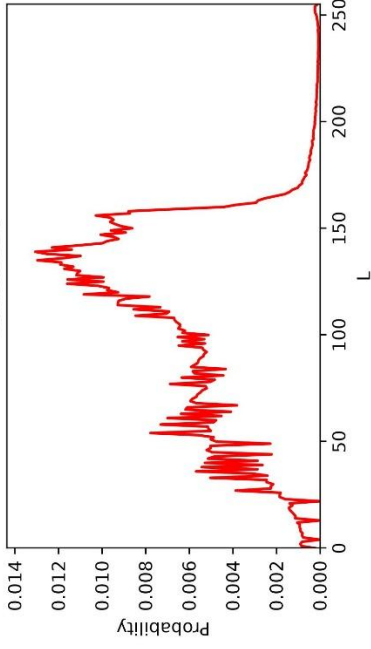
# Calculate output PDF and CDF
output_hist = np.bincount(output_L.ravel(), minlength=256)
output_pdf = output_hist / output_L.size
output_cdf = np.cumsum(output_pdf)

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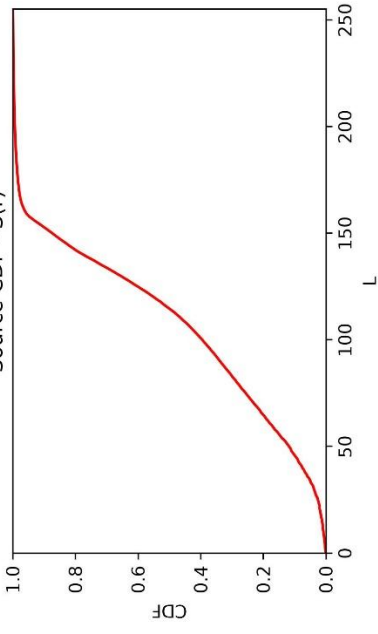
Input Image



Source PDF



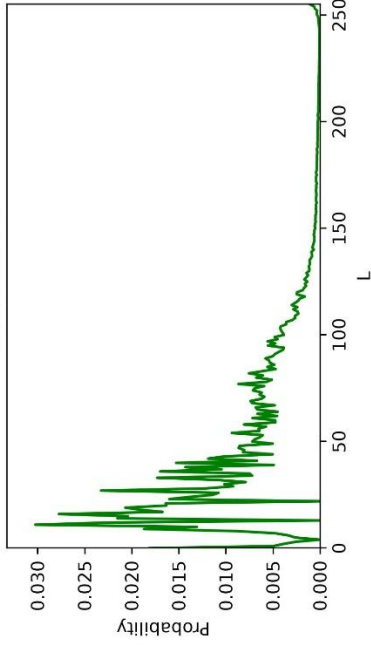
Source CDF - $S(r)$



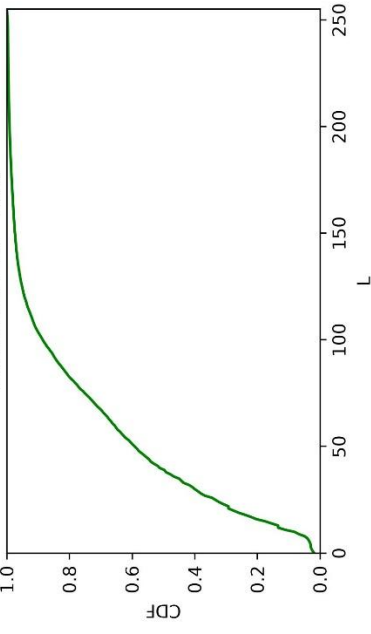
Reference Image



Reference PDF



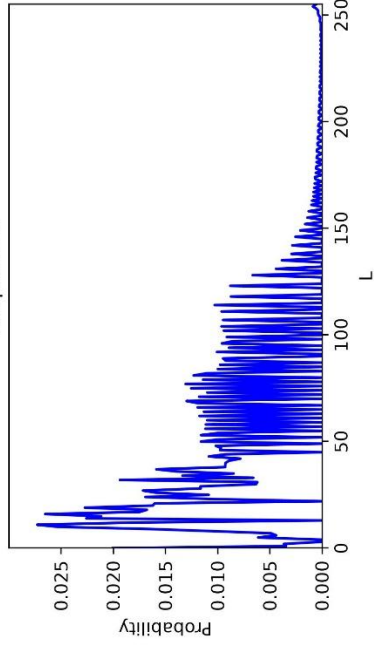
Reference CDF - $G(z)$



Output Image



Output PDF



Output CDF

